Ground-Roll Attenuation Using a Dual-Filter-Bank Convolutional Neural Network

Chao Zhang and Mirko van der Baan

Abstract—**Ground-roll attenuation is very challenging because of its high amplitudes and overlapping frequency content with desired signals. A particular challenge is to recover weak reflections underneath strong masking ground-roll. We propose a dual-filter bank setup combined with two convolutional neural networks (CNNs) to realize ground-roll attenuation. The rationale for using a dual-filter bank strategy is that it permits using two CNNs with different input kernel sizes and different complexities to recognize and extract broad-scale (low-wavelength) and narrow-scale (high-wavelength) features separately. We also apply a frequency filter to create a preliminary separation between the signal and the noise. In addition, we use a radial trace transform that focuses desired signal to a smaller area, facilitating separation of the reflections and ground-roll and accelerating training. The network training strategy combines synthetic and field data examples, in addition to noise injection to augment the number of available training samples. Tests on synthetic and field datasets show that the proposed strategy achieves superior ground-roll attenuation compared with standard methods, even in the case of data with irregular spatial spacing or ground-roll characteristics not contained in the training data.**

Index Terms—**Convolutional neural network (CNN), ground-roll attenuation, noise suppression, radial trace (RT) transform, signal recovery.**

I. Introduction

GROUND-ROLL is observed during acquisition of seismic reflection data. It is composed of surface waves, masking the underlying reflections. Separation of ground-roll and reflections is challenging since the ground-roll has often substantially stronger amplitudes, and overlap occurs in the temporal, spatial, and spectral domains.

A variety of methods have been developed for ground-roll attenuation in the processing stage. The basic principle is to find a domain in which the signal and noise are well separated, so that the signal is not affected after ground-roll removal. Common solutions are to estimate and remove ground-roll in the frequency–wave number (f–k) domain [1], localized frequency transform domain [2], radial trace (RT) domain [3]–[6], wavelet or directional wavelet domain [7]–[13], or by learning an appropriate basis [14]. These solutions use the principle that surface waves are characterized by both lower frequencies and lower apparent moveout velocities than reflections, which leads to clear separation in the f–k domain, or their directional-wavelet equivalents, if and only if the data are spatially sufficiently densely sampled. These methods are simple and fast by design, since ground-roll attenuation is performed in the early processing stages, thus dealing with large data volumes. Unfortunately, spatial aliasing is common, strongly reducing the efficiency of standard separation techniques [15]. An alternative approach is to do forward modeling of the ground-roll, creating a prediction–subtraction strategy [16]–[18]. Unfortunately, this strategy requires accurate knowledge of the medium properties, creating first an additional inverse problem.

Recently, deep learning methods have been investigated for ground-roll attenuation with the availability of large amounts of data and significant computational power, such as convolutional neural network (CNN) [19]–[21] and generative adversarial network [22]–[24]. These methods prove that deep learning can achieve a comparable result with traditional methods but with an increased efficiency. The advantages of fully trained deep learning algorithms are that there are no needs to set multiple tuning parameters and specify any prior information on signal or noise characteristics to obtain suitable results. But when the energy of ground-roll is much stronger than reflection events or the region spanned by ground-roll is larger, several challenges arise for the deep learning algorithms.

1) The weak events underneath the ground-roll may not be well reconstructed because of their overlap in the spatial, temporal, and spectral domains, rendering their identification and recovery difficult.
2) There are often some short spurious events generated by the network as some lateral correlated components of the ground-roll are identified as reflection events by the network.
3) It is difficult for networks with a single convolutional kernel size to recognize both broad- and narrow-scale features.

In this article, we propose a dual-filter bank CNN (DFB-CNN) to address the issues mentioned above. First, we design two networks with different complexities to extract broad- and narrow-scale features separately. The CNN constructed for the broad-scale component has a larger kernel size and deeper layers compared with the one used for narrow-scale feature extraction. In this way, low-frequency ground-roll is easily recognized and suppressed, and high-frequency signal is better preserved.

In addition, we add an RT transform to compact ground-roll and reflections, thereby emphasizing their differences in terms of apparent velocity and spatial geometric features, facilitating

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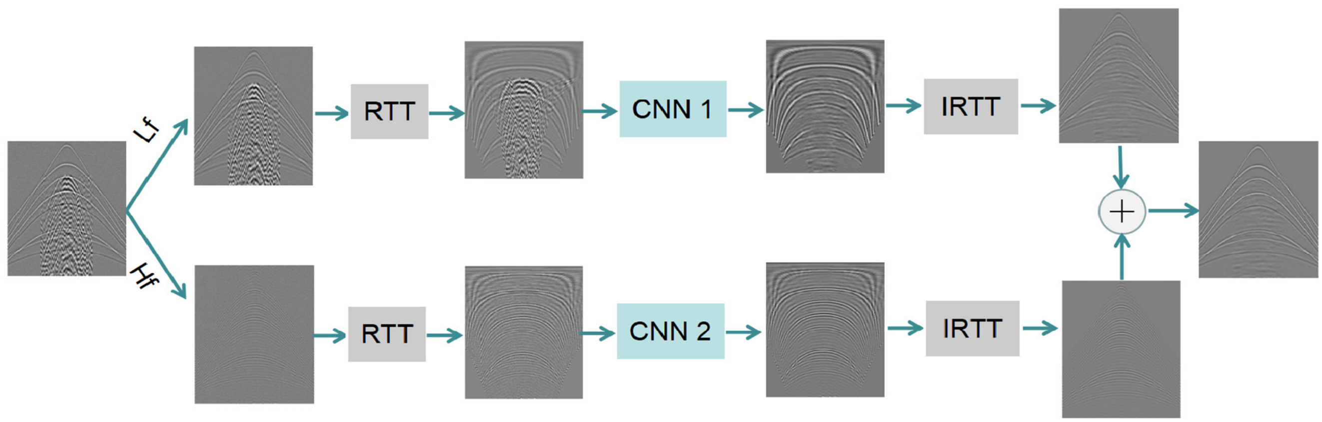

Fig. 1. Structure of the DFB-CNN. Lf and Hf represent the low- and high-frequency separation. RTT and IRTT represent, respectively, the forward and inverse RT transforms. CNN1 and CNN2 are two CNNs with different network architectures aimed at recognizing and extracting, respectively, broad- and narrow-scale features.

ground-roll identification by the dual CNNs. Furthermore, the frequency content of ground-roll becomes lower in the RT domain since the RT sample trajectories are nearly parallel to the wavefronts of ground-roll, further improving network performance.

The rest of this article is organized as follows. In Section II, we first describe the basic theory of the RT transform, the architecture of the DFB-CNN, training strategy, including training data. Experimental results on synthetic and real datasets are shown in Section III. Section IV contains the discussion, and the main conclusions are summarized in Section V.

II. Methodology

A. Dual-Filter Bank Convolutional Neural Network

We design two parallel networks with different network complexities to, respectively, recover low- and high-frequency signals in an end-to-end manner. In addition, the RT transform and its inverse form are added before and after the CNN to reduce the overlapping contents in both time and frequency domains, allowing for enhanced feature recognition and extraction. The structure of the proposed DFB-CNN is shown in Fig. 1.

Consider noisy seismic data $\mathbf{y}$ which are comprised of signal $\mathbf{s}$ and noise $\mathbf{n}$, that is

$$\mathbf{y} = \mathbf{s} + \mathbf{n}. \tag{1}$$

The noisy seismic data $\mathbf{y}$ is divided into a low-frequency component $\mathbf{y}_l$ by a low-pass filter and a high-frequency component $\mathbf{y}_h$, where $\mathbf{y}_h = \mathbf{y} - \mathbf{y}_l$. $\hat{\mathbf{s}}_l$ and $\hat{\mathbf{s}}_h$ are denoised versions of low-frequency component $\mathbf{y}_l$ and high-frequency component $\mathbf{y}_h$ by the proposed method. They are described as

$$\hat{\mathbf{s}}_l = \mathbf{R}^{-1}\left\{\mathbf{R}\{\mathbf{y}_l\} - F_{\Theta_1}(\mathbf{R}\{\mathbf{y}_l\})\right\} \tag{2}$$

$$\hat{\mathbf{s}}_h = \mathbf{R}^{-1}\left\{\mathbf{R}\{\mathbf{y}_h\} - F_{\Theta_2}(\mathbf{R}\{\mathbf{y}_h\})\right\} \tag{3}$$

where $\mathbf{R}$ and $\mathbf{R}^{-1}$ represent the RT transform and its inverse transform. $F_{\Theta_1}(\cdot)$ and $F_{\Theta_2}(\cdot)$ represent the networks trained for low- and high-frequency components, respectively. Θ_1 and Θ_2 are the corresponding network parameters, weights, and biases. The networks predict the noise instead of the signal (Fig. 2) because this strategy has been demonstrated to perform better with reduced training times and smaller architectures [25].

Finally, the clean signal estimation $\hat{\mathbf{s}}$ is obtained by

$$\hat{\mathbf{s}} = \hat{\mathbf{s}}_l + \hat{\mathbf{s}}_h. \tag{4}$$

The mapping functions $F_{\Theta_1}(\cdot)$ and $F_{\Theta_2}(\cdot)$ in the proposed framework can be any network. In this article, a classic CNN setup with ReLU activation functions and batch normalization is used; yet, the networks are trained to predict the noise instead of the signal [25]. Its architecture is shown in Fig. 2.

The architecture has three types of layers represented by three colors. The convolutions in each layer are described by a 4-D tensor [26]. Between the input layer and the first hidden layer, the convolutional kernels have a size of $k \times k \times n_{f,1}$, where k is the size of convolutional kernels in the horizontal and vertical directions and $n_{f,1}$ is the number of feature maps in the first hidden layer. For each subsequent hidden layer j, the convolutional kernels have a size of $k \times k \times n_{f,j} \times n_{f,j+1}$, each feature map is created by a new convolutional filter, so the last dimension corresponds in essence to the number of convolutional filters.

ReLU is an activation function which leaves all positive numbers unchanged but renders all negative numbers zero [27]. Batch normalization [28] is added between convolutions and ReLU activation functions starting from the second layer. It normalizes the output of a previous layer by subtracting the batch mean and dividing by the batch standard deviation. Batch normalization is a way to increase the learning speed and reduce sensitivity to initialization. The last convolution layer is used to reconstruct the output from the feature map. This network is used to train a residual mapping to predict the error. Finally, the signal can be obtained by the difference between the noisy data and the predicted error.

The CNN constructed for the low-frequency component has a larger kernel size. Also this network has more layers and feature maps because ground-roll is concentrated in this

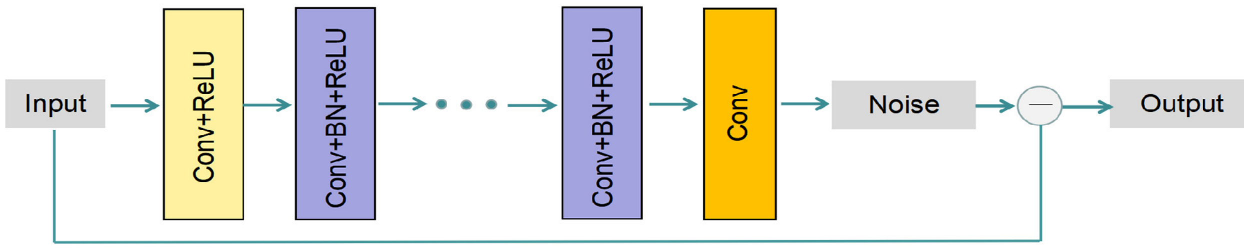

Fig. 2. Structure of CNN used in Fig. 1 [25]. The architecture has three types of layers represented by three colors. Batch normalization is added between convolution and ReLU activation function starting from the second layer. This network is used to train a mapping to predict the noise. The signal is obtained from the difference between the input data and the predicted noise.

low-frequency component, requiring a stronger feature recognition ability.

The specific network parameters are set as follows.

We use a nine-layer network for low-frequency signal reconstruction. There are 100 filters of size 5×5 for the first layer to generate 100 feature filters of size $5 \times 5 \times 100$ for the second to eighth layers, followed by one filter of size $5 \times 5 \times 100$ for the last layer. There are 1.75 million free parameters in total. A five-layer network is adopted for high-frequency signal reconstruction. There are 64 filters of size 3×3 for the first layer to generate 64 feature filters of size $3 \times 3 \times 64$ for the second to fourth layers, followed by one filter of size $3 \times 3 \times 64$ for the last layer. This network contains 0.1 million free parameters.

B. Radial Trace Transform

The RT transform maps the amplitudes of seismic trace $s(x, t)$ from the distance–time domain to the velocity–time domain $s'(v, t')$ using a fan of constant-velocity trajectories radiating from a common origin [3], [4]. The RT transform $\mathbf{R}$ is defined as

$$\mathbf{R}\{s(x,t)\} = s'\left(v, t'\right) \tag{5}$$

the inverse transform is given by

$$\mathbf{R}^{-1}\left\{s'\left(v, t'\right)\right\} = s(x,t) \tag{6}$$

where

$$t' = t - t_0, \quad v = \frac{x - x_0}{t - t_0}. \tag{7}$$

Here, x is the offset and t is the two-way traveltime. After the RT transform, the new coordinates become apparent velocity v and shifted two-way traveltime t'. x_0 and t_0 are the origin position of the RT transform. The choice of origin position depends on the apparent velocity of the coherent noise to be suppressed. For ground-roll attenuation, it can be chosen at the source point of the shot gather in the $x - t$ domain.

The process of mapping a seismic shot gather from the $x - t$ domain to the RT domain is shown in Figs. 3 and 4 [4]. Fig. 3 illustrates how the RTs are sampled from the $x - t$ shot gather. A fan of RT trajectories (dashed lines) representing a wide range of velocities is used to extract samples (seismic amplitudes) along each of the linear trajectories from the $x - t$ domain seismic traces, then the samples gathered along each dashed line form a single trace in the RT domain.

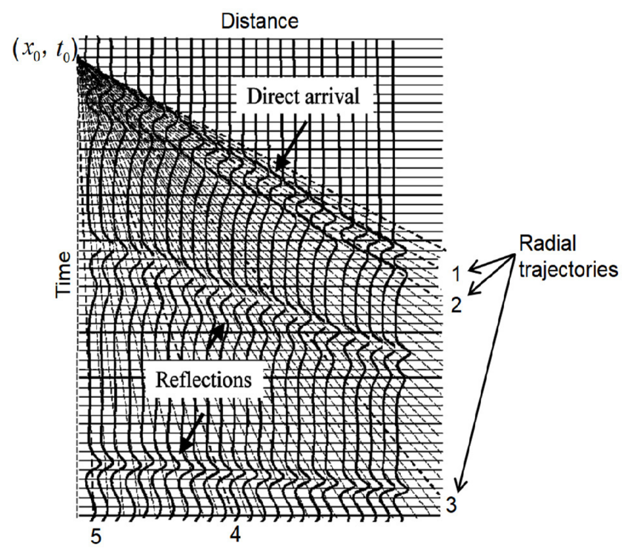

Fig. 3. RT trajectories superimposed on an $x - t$ panel of seismic traces. A fan of RT trajectories (dashed line) representing a wide range of velocities is used to map an $x - t$ panel to the RT domain. Adapted from [4].

The RTs extracted from the $x - t$ panel in Fig. 3 along the numbered trajectories are shown in Fig. 4. The waveform is stretched when the angle between the event wavefront and radial trajectory in the $x - t$ domain is small, as shown by radial trace 1 in Fig. 4. This event stretch decreases the apparent frequency of the event in the RT domain. When the angle between the event wavefront and radial trajectory in the $x - t$ domain becomes larger, consequently the waveform is less stretched in the RT domain, as shown by radial traces 2–5 in Fig. 4. The radial trajectories usually cross the reflection events at large angles since the reflection events do not radiate from the origin of the radial trajectories, so the waveforms and apparent frequency of reflection events are nearly unchanged.

Since the radial trajectories radiate from a common origin, they match the pattern of much coherent noise radiating away from the source point in the $x - t$ domain. The RT transform can capture source-generated noises of any velocity by choosing an appropriate range of velocities and the radiating origin. The coherent noise such as ground-roll will be isolated into relatively small groups of traces near the center of the gather in the RT domain because it has low apparent velocity and passes through the RT origin (Fig. 5). Reflection events which do not pass through the RT origin will remain spread laterally across the RT domain.

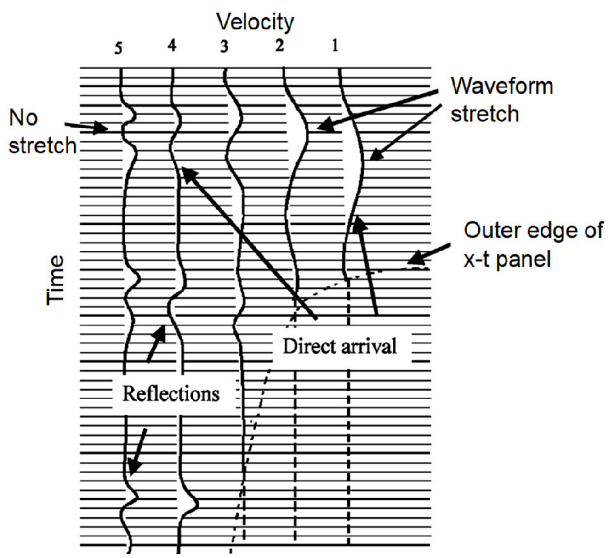

Fig. 4. Representative numbered RTs extracted from the $x-t$ panel in Fig. 3. The waveforms on traces 1 and 2 are stretched as these two traces are nearly parallel to the first event in Fig. 3. The mapped outer edge of the $x-t$ panel is represented by the dashed contour. Adapted from [4].

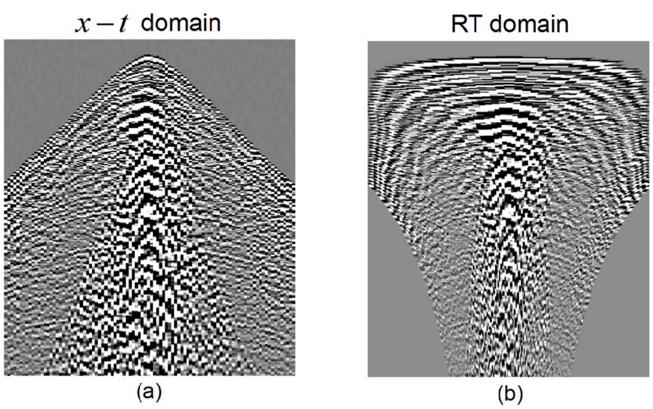

Fig. 5. (a) Raw shot gather in the $x-t$ domain. (The horizontal axis represents offset with the range from −960 to 960 m; the vertical axis represents time with the range from 0 to 2.86 s.) (b) Shot gather of (a) in the RT domain. (The horizontal axis represents velocity with the range from −3200 to 3200 m/s; the vertical axis represents time with the range from 0 to 2.86 s.) Ground-roll is isolated into a relatively small area near the center of the gather in the RT domain.

The RT transform increases the visibility of the target reflections and facilitates their subsequent detection and extraction. In addition, the RT transform makes it easier to deal with aliased ground-roll or missing traces, again because of signal focusing.

C. Network Optimization

Finding the optimum network parameters Θ_1 and Θ_2 in (2) and (3) is done by formulating training as a minimization problem, that is

$$\arg\min_{\Theta_1} \frac{1}{N}\sum_{i=1}^{N}\left\|F_{\Theta_1}\left(\mathbf{R}\left\{y_{l,i}\right\}\right), \mathbf{R}\{n_{l,i}\}\right\|_F^2 \tag{8}$$

$$\arg\min_{\Theta_2} \frac{1}{N}\sum_{i=1}^{N}\left\|F_{\Theta_2}\left(\mathbf{R}\left\{y_{h,i}\right\}\right), \mathbf{R}\{n_{h,i}\}\right\|_F^2 \tag{9}$$

where the low-frequency noisy data $y_{l,i}$ and low-frequency noise $n_{l,i}$ represent the input and output training pairs. $y_{h,i}$ and $n_{h,i}$ are the corresponding high-frequency training pairs. i is the index of each training pair, and N is the total number of training pairs. $\|\cdot\|_F$ represents the Frobenious norm. In this article, we use the mean square error (MSE) as a loss function to evaluate the misfit between the actual and desired output.

The network parameters Θ_1 and Θ_2 are trained by back-propagation [29]. To economize on the computational cost, a mini-batch stochastic gradient descent [30] is used for local gradient calculation. In every iteration, not all training samples are required in the network optimization instead random subsets (called batches) of the whole training samples are used, followed by computing the average gradient since this may substantially accelerate convergence [27]. Batch training leads to a reduced computation cost compared to using the whole dataset, but it still provides a good approximation to the true gradient. In this article, the batch size is set as 32.

We use adaptive moment estimation (Adam) [31] with default parameters to gradually update and optimize the network parameters Θ_1 and Θ_2. The Adam optimizer leverages the power of an adaptive learning rate method to find individual learning rates for different parameters, which can achieve a better training performance.

D. Training Set Construction

Network performance is tied to the completeness and representativeness of the provided training datasets [32]. Typically training consists of providing sets of input–target pairs that are noisy–clean data. We replace them as noisy–noise data pairs as the residual mapping learning converges faster and achieves a better reconstruction performance compared with networks trained to predict the clean signal [25]. In addition, we use noise injection in that we purposely add random noise to the input for data augmentation purposes and improved noise robustness [32].

We use a combination of synthetic and field data as training examples. Synthetic data are adopted because it is difficult to acquire clean signal from the field records. Preprocessing can be done to get noise-reduced data for training, but as a result, deep learning methods are unlikely to surpass the quality of the preprocessing training examples, leading to results that could have been obtained by applying the preprocessing directly to the field dataset. Furthermore, the reflection events entangled with ground-roll have typical hyperbolic features, which can be well simulated by synthetic data. Conversely, field examples of surface waves are added since these signals are characterized by a wide variety of features, which are difficult to model, while ensuring both their completeness and full representation.

We create 100 noiseless shot gathers of 128 traces and 1500 time samples with hyperbolic moveout curves using Ricker wavelets. For increased variability, the Ricker wavelets can be 0, minimum or maximum phase. Each shot gather comprises 6–20 reflections, starting at random zero-offset arrival times. Each hyperbolic reflection has its own moveout velocity, with its apex at zero offset. Details on the parameter settings of the synthetic datasets are shown in Table I.

TABLE I
PARAMETER SETTINGS OF THE SYNTHETIC DATASETS

Parameter	Specification
Wavelet	Ricker wavelets
Dominant frequency (Hz)	15-30
Velocity (m/s)	600-3000
Amplitude	-1-1
Sample interval (ms)	2
Receiver interval (m)	20

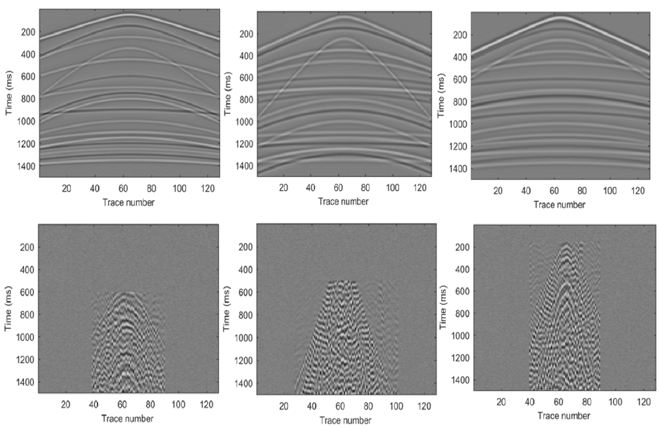

Fig. 6. (Top row) Three representative signal and (bottom row) noise datasets used for training.

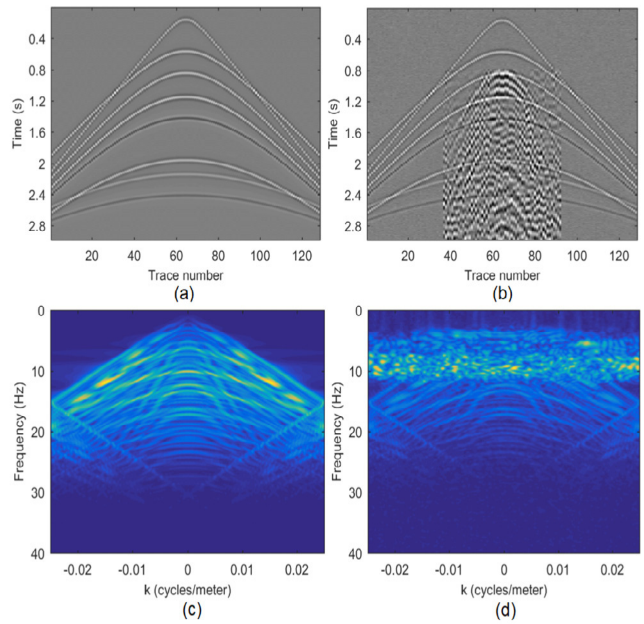

Fig. 7. Synthetic records and the corresponding f–k spectra. (a) Noise-free data. (b) Data contaminated with ground-roll and random noise. (c) and (d) f–k spectra corresponding to (a) and (b). The reflections are spatially aliased as seen by their wrap-around effects in the f–k domain. The ground-roll strongly overlaps with the reflections below 10 Hz.

Then, we randomly added ground-roll extracted from land seismic datasets with different amplitudes to the noiseless synthetic gathers. In 3-D acquisitions, surface waves may not originate from the origin, but may have a hyperbolic appearance starting at later times. Hence, we include both surface waves from in-line and cross-line acquisitions. The extracted ground-roll has sometimes its head and/or sides partially removed (Fig. 6, bottom row), which we permitted since it adds another layer of complexity to the training process. To increase the generalization of the network and the ability to suppress random noise, white Gaussian noise with the standard deviation (0–0.2) is also added to the noiseless synthetic gathers. Some representative signal and noise datasets are shown in Fig. 6.

Next, we divide the signal and noise datasets into low- and high-frequency components. The low-frequency components are obtained by a low-pass filter with the range from 0 to L Hz. The high-frequency components are the difference between the inputs and the low-frequency components. Frequency L is randomly chosen from a range. This is also an effective data augmentation strategy since L is not a fixed value and leads to more robust results at both the application and training stage, since the networks are not trained for specific thresholds which may be suboptimal for unseen field data. In this article, the range of L is chosen from 10 to 20 Hz. Then, all these low- and high-frequency components of the signal and noise are transformed into the RT domain by the RT transform. The velocity range of RTs is between -3200 and 3200 m/s. The velocity sample increment is 6 m/s. We crop the data in the RT domain into small patches. Training patches are generated using a sliding window of size 80×80 with stride 36. Finally, 64 960 signal and noise examples for respective low- and high-frequency components are generated for training. The batch size for training is 32. We set an exponential decay learning rate from 10^{-3} to 10^{-5} for the 50 epochs. We use the MatConvNet package [33] to train the proposed models. The experiments are carried out in the MATLAB (R2016b) environment running on a PC with Intel Core i9-10900X CPU and an Nvidia GeForce RTX 2080 GPU. The training process took a total of 12 h 46 min and 3 h 50 min for, respectively, the low- and high-frequency components.

III. Numerical Tests

A. Synthetic Data Processing

The performance of the proposed method is tested on a synthetic example with similar characteristics to the training data [Fig. 7(a)]. This test is to validate training completed properly. The sample interval is 2 ms. The trace spacing is 20 m. A Ricker wavelet is used to simulate the synthetic seismic signal. The noisy data in Fig. 7(b) are generated by adding ground-roll extracted from a raw shot gather and white Gaussian noise to the noise-free data [Fig. 7(a)], yielding an S/N ratio of 0.4152 dB. The signal-to-noise ratio is defined by the power ratio [34]. The ground-roll overlaps with the refection events in both time and frequency domains [Fig. 7(b) and (d)].

To better demonstrate the superiority of the proposed method, we compare it with a classic ground-roll removal method (f–k filtering) and a single CNN specifically trained for ground-roll attenuation [25]. The apparent moveout velocity for f–k filtering is between -650 and 650 m/s. For the

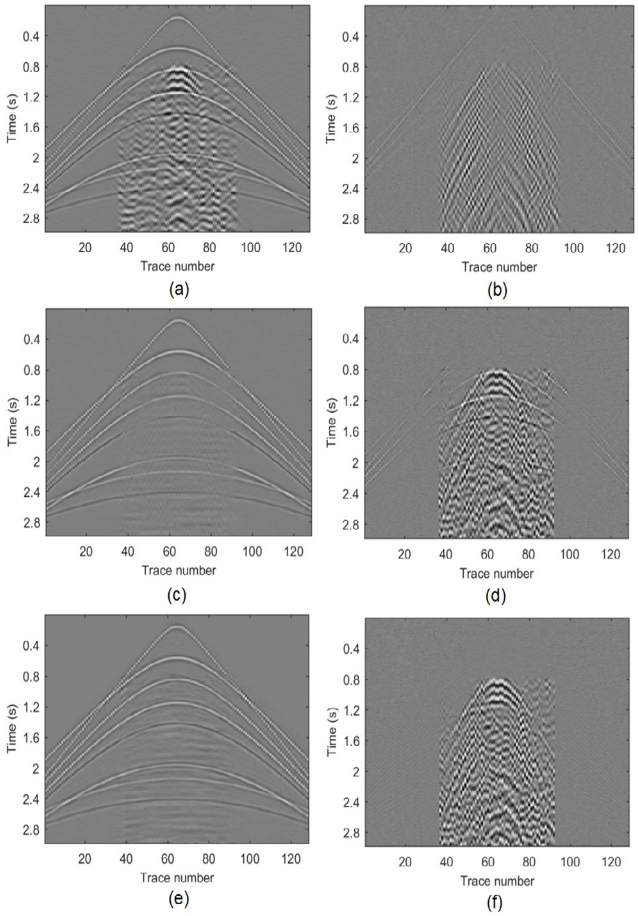

Fig. 8. Ground-roll removal results and difference profiles of Fig. 7(b). Results after (a) f–k filtering, (c) single CNN, and (e) DFB-CNN. Difference profiles for (b) f–k filtering, (d) single CNN, and (f) DFB-CNN.

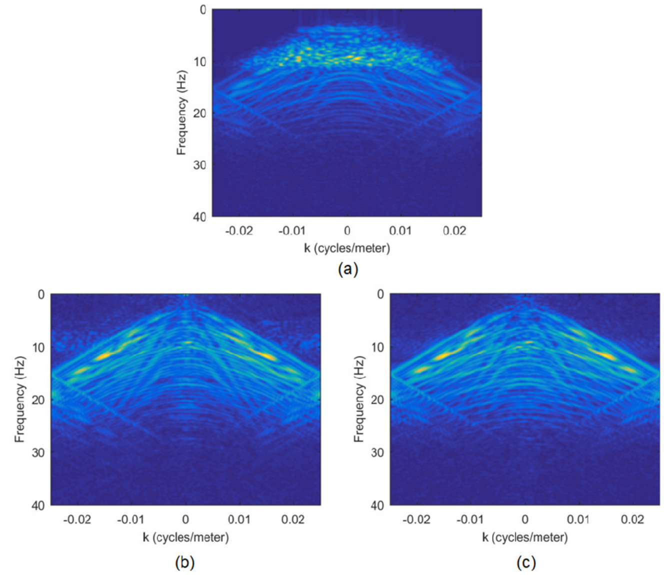

Fig. 9. f–k spectra of the ground-roll removal results in Fig. 8 by different methods. The f–k spectra after (a) f–k filtering, (b) single CNN, and (c) DFB-CNN. Compare with Fig. 7(c) and (d).

TABLE II

PERFORMANCE STATISTICS FOR DIFFERENT FILTERING TECHNIQUES

Method	F-k filtering	Single CNN	DFB-CNN
S/N ratio (dB)	4.0080	8.9488	10.7455
RMSE	0.1059	0.0484	0.0376
computation time (s)	0.0153	0.0526	0.7669

DFB-CNN, the demarcation point L between low- and high-frequency components is 15 Hz. This value is chosen because ground-roll usually concentrates in 0–15 Hz. Changing the value of separation frequency L by a few Hz has no significant impact on the separation results in this test due to its randomization during training. The network structure for the single CNN used for comparison is shown in Fig. 2. To guarantee the network performance of one single CNN, the network parameters are set as suggested in [25], a 17-layer network is adopted for noise removal. There are 64 filters of size 3×3 for the first layer to generate 64 feature filters of size $3 \times 3 \times 64$ for the 2nd–16th layers, followed by one filter of size $3 \times 3 \times 64$ for the last layer. There are 0.56 million free parameters in total. The signal and noise training datasets are the same as the ones used in the proposed DFB-CNN. But there is no frequency division or RT transform module. Training patches are generated directly from the datasets in the time-space domain using a sliding window of size 80×80 with stride 15. In total, 70 208 signals and noise examples are generated for training. The batch size for training is 32. We set an exponential decay learning rate from 10^{-3} to 10^{-5} for the 50 epochs.

The ground-roll removal results and difference profiles of Fig. 7(b) by f–k filtering, single CNN, and the proposed DFB-CNN are shown in Fig. 8. f–k filtering cannot deal well with the ground-roll which has an overlapping frequency and wavenumber content with the reflections [Fig. 8(a) and (b)]. The single CNN deals better with the overlapping frequency content [Fig. 8(c) and (d)] but some high-frequency reflections are also removed. In addition, the reflections underneath the surface waves are not well recovered either.

In comparison, the proposed DFB-CNN performs best. High-frequency reflections are well preserved as are the weak reflections hidden by the ground-roll [Fig. 8(e) and (f)]. This test demonstrates that the proposed dual-filter bank network is more appropriate for this type of application, producing smaller architectures with enhanced detection of both narrow- and broad-scale features.

The f–k spectra of the processing results in Fig. 8 are shown in Fig. 9. Low-frequency components of reflection events are still entangled with ground-roll after f–k filtering [Fig. 9(a)]. The ground-roll is well suppressed by both the single CNN and DFB-CNN [Fig. 9(b) and (c)], but DFB-CNN makes the best trade-off between signal preservation and ground-roll elimination as seen from its f–k spectrum [Fig. 9(c)].

Table II presents a comparison of these three methods in the S/N ratio, root MSE (RMSE), and computation time. Larger

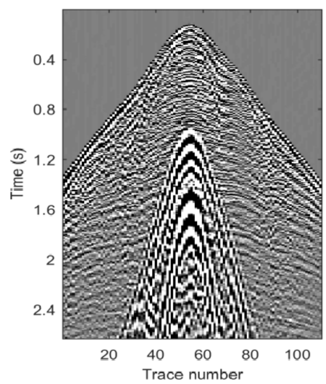

Fig. 10. Common shot gather contaminated by high-energy ground-roll.

S/N ratios and lower RMSE indicate better processing results, demonstrating a better performance of the proposed DRT-CNN over the single CNN and f–k filtering. Computation times on a single-shot gather are provided for illustration purposes. f–k filtering is CPU-optimized, the single CNN is GPU-optimized, whereas the DFB-CNN uses both CPU and GPU optimized modules. f–k filtering has the lowest computational cost. The single CNN has an attractive computation time, partially due to its GPU optimization. The DFB-CNN is an order of magnitude slower, largely because of the additional forward and inverse RT transforms which is CPU optimized.

B. Field Data Processing

We first investigate the performance of the proposed method on a common shot seismic gather from northern China (Fig. 10). This shot gather is excluded from the training data. The sample interval of this dataset is 2 ms. The trace spacing is 20 m. The ground-roll in this shot gather arrives late since the source is offset from the receiver line, and backscattered energy is visible at later times. Again, the trained DFB-CNN is compared with f–k filtering and the trained single CNN. The network structure, training dataset, and training parameters of the proposed DFB-CNN are as described in the theory section. No additional training was performed for either network. The parameter setting of f–k filtering and the frequency demarcation point of the DFB-CNN are the same as the ones used in the synthetic example.

The ground-roll attenuation results and difference profiles of f–k filtering, single CNN, and the proposed DFB-CNN are shown in Fig. 11. The corresponding f–k spectra are shown in Fig. 12. The direct and refracted waves are well preserved by f–k filtering compared with the CNN-based methods, but residual ground-roll still blocks the underlying reflection events [Fig. 11(b)]. Most ground-roll has been attenuated by the single CNN although some residual imprint remains [Fig. 11(c)]. In addition, some reflections are removed in the far offsets as seen from the difference profile and the f–k spectrum of the single CNN [Figs. 11(d) and 12(c)].

In comparison, the proposed DFB-CNN performs best for both high-energy ground-roll suppression and signal preservation due to the dual-filter bank strategy [Fig. 11(e) and (f)]. This can also be seen from their f–k spectra comparison

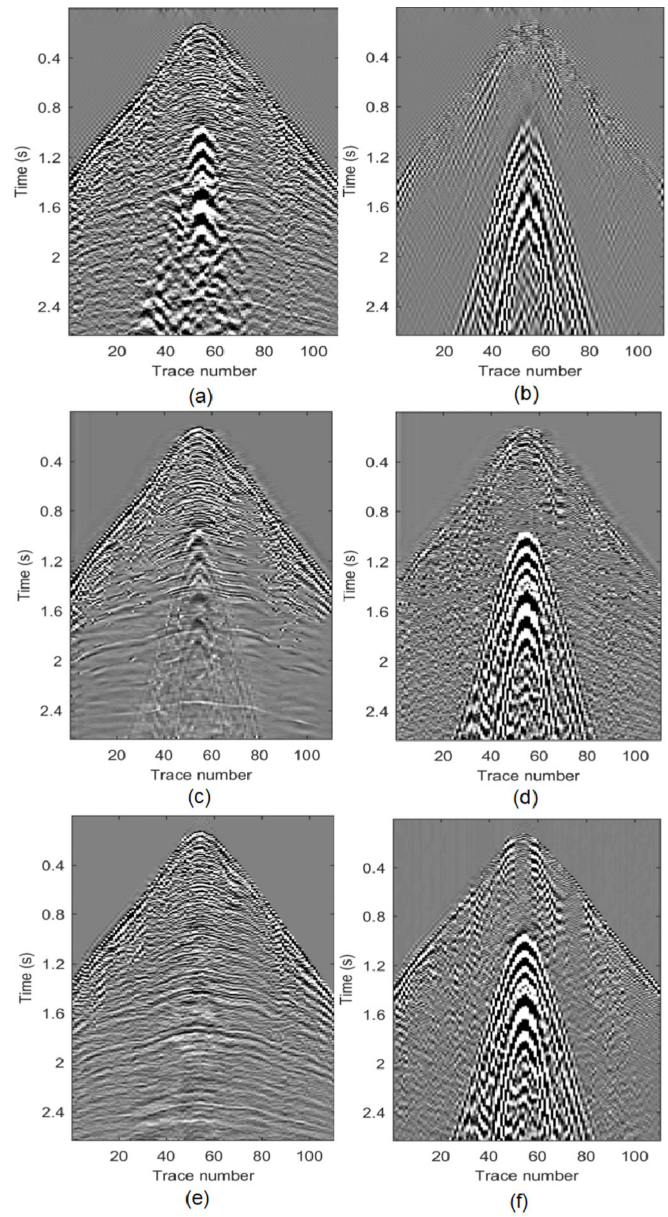

Fig. 11. Ground-roll removal results and difference profiles of Fig. 10. Results after (a) f–k filtering, (c) single CNN, and (e) DFB-CNN. Difference profiles for (b) f–k filtering, (d) single CNN, and (f) DFB-CNN.

in Fig. 12. The reflection events, especially the events underneath ground-roll, are well recovered by the proposed DFB-CNN, which is difficult to achieve for most traditional and deep learning-based ground-roll attenuation methods. Deep learning methods can also remove random noise since the white Gaussian noise is added in the training processing [Fig. 11(d) and (f)]. Although the removed random noise is less similar to white Gaussian noise, the network learns to remove random features which demonstrates the network we trained has a higher generalization [32].

Finally, we test the proposed DFB-CNN on another common shot gather (also excluded from the training data) contaminated by high-amplitude irregular ground-roll (Fig. 13). This dataset is also collected in northern China. The sample interval of this dataset is 2 ms. The trace spacing is 20 m.

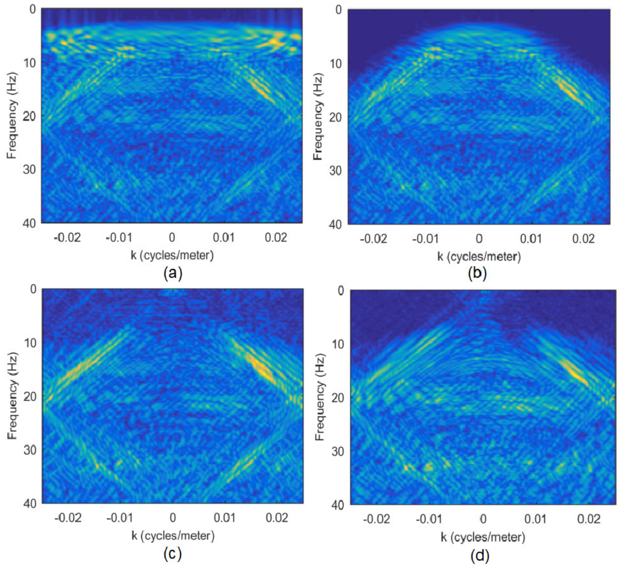

Fig. 12. f–k spectra of the common shot gather (Fig. 10) and the ground-roll removal results in Fig. 11 by different methods. (a) f–k spectrum of the common shot gather. f–k spectra after, (b) f–k filtering, (c) single CNN, and (d) DFB-CNN.

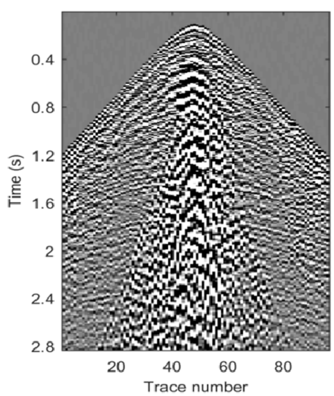

Fig. 13. Second common shot gather contaminated by aliased and irregular ground-roll.

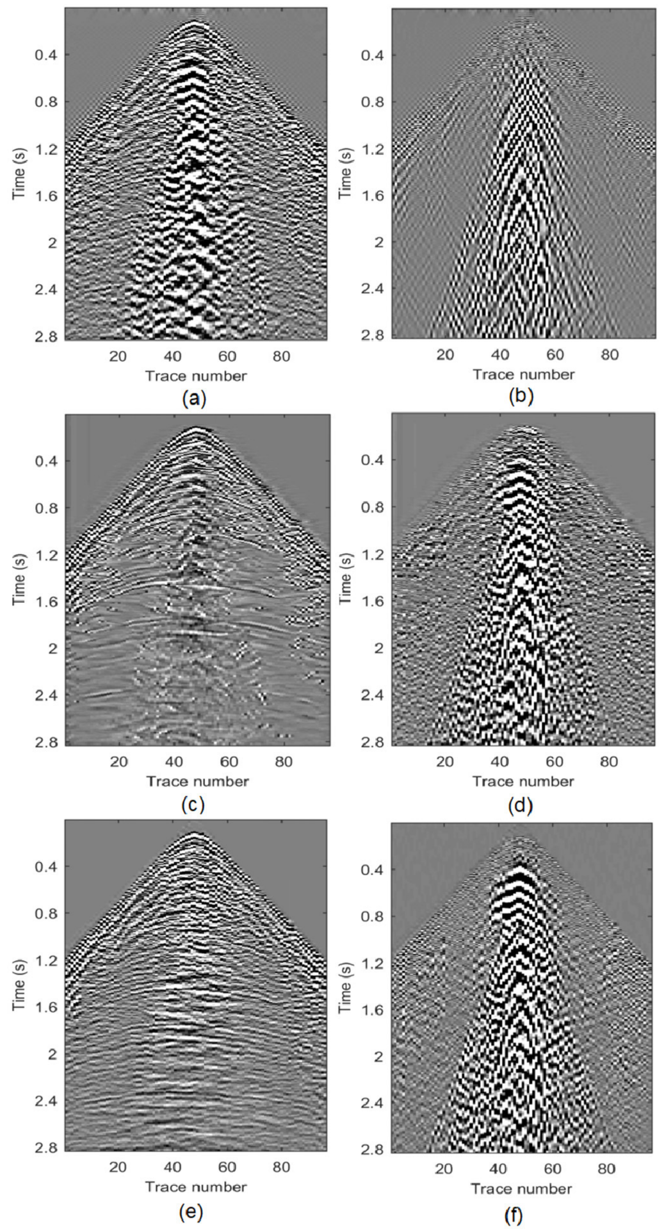

Fig. 14. Ground-roll removal results and difference profiles of Fig. 13. Results after (a) f–k filtering, (c) single CNN, and (e) DFB-CNN. Difference profiles for (b) f–k filtering, (d) single CNN, and (f) DFB-CNN.

This dataset contains strong, irregular, and aliased ground-roll. The continuity of the reflection events is greatly disrupted and less continuous than in the previous example. The parameter setting of f–k filtering and the frequency demarcation point of the DFB-CNN are unchanged.

Fig. 14 shows the ground-roll removal results and difference profiles of f–k filtering, the single CNN, and the proposed DFB-CNN. Corresponding f–k spectra are shown in Fig. 15. f–k filtering performs poorly due to the strongly irregular ground-roll and the presence of both spatially aliased reflections and ground-roll. The CNN approaches yield better results (Figs. 14 and 15). There are still some residual ground-roll and artifacts after processing by the single CNN. In addition, the reflection events are over-smoothed and the high-frequency components of the events are not well reconstructed [Figs. 14(c) and (d) and 15(c)]. The results confirm again that a single CNN performs less well when the target signals and noise have overlapping broad- and narrow-scale features.

The reflection events are well reconstructed and the continuity of the events are enhanced by the proposed DFB-CNN due to the dual-filter bank design [Figs. 14(e) and (f) and 15(d)]. The RT transform aids in the separation as well. The proposed DFB-CNN can remove some random or correlated noise which are not included in the training set, which indicates a higher generalization of the proposed network.

Proper generalization is a key criterion to evaluate the performance of the trained networks. The previous field data examples have demonstrated that the proposed network works well for high-energy ground-roll attenuation, random noise suppression, and reconstruction of reflections. Yet, it would be undesirable if the network has to be retrained when the trace spacing changes. As a stringent test of

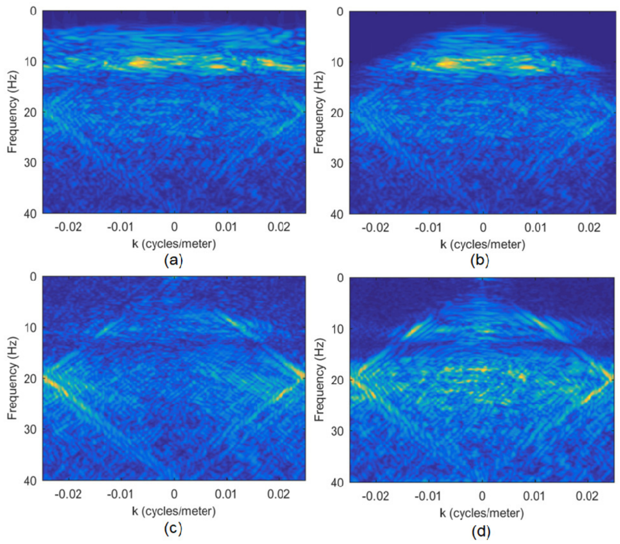

Fig. 15. f–k spectra of the common shot gather (Fig. 13) and the ground-roll removal results in Fig. 14 by different methods. (a) f–k spectrum of the common shot gather. The f–k spectra after (b) f–k filtering, (c) single CNN, and (d) DFB-CNN.

appropriate network generalization, we remove every other trace in Fig. 13, that is, a trace reduction of 50%. This is a challenging test since it doubles trace spacing, weakening lateral signal coherency, simultaneously strongly increasing the spatial aliasing of reflections and ground-roll, and halving the Nyquist wavenumber.

Figs. 16(a) and 17(a) show the resulting shot gather and f–k spectrum. The geometric structure of the ground-roll changes and the reflection events are difficult to identify after 50% traces are removed [Figs. 16(a) and 17(a)]. Unsurprisingly, f–k filtering fails to suppress the ground-roll [Figs. 16(b) and 17(b)]. The single CNN removes the ground-roll but does not recover the reflections well [Figs. 16(c) and 17(c)]. The proposed DFB-CNN still successfully attenuate the ground-roll and the reflection events underneath the ground-roll are reasonably well reconstructed although some of the extracted horizontal, low-frequency arrivals near trace 25 (between 1.2 and 2.5 s) are likely ground-roll artifacts [Figs. 16(d) and 17(d)]. Nonetheless, this challenging test demonstrates that the dual-filter bank networks are capable of recognizing ground-roll and reflections even if the trace spacing is significantly different from that used in the training stage.

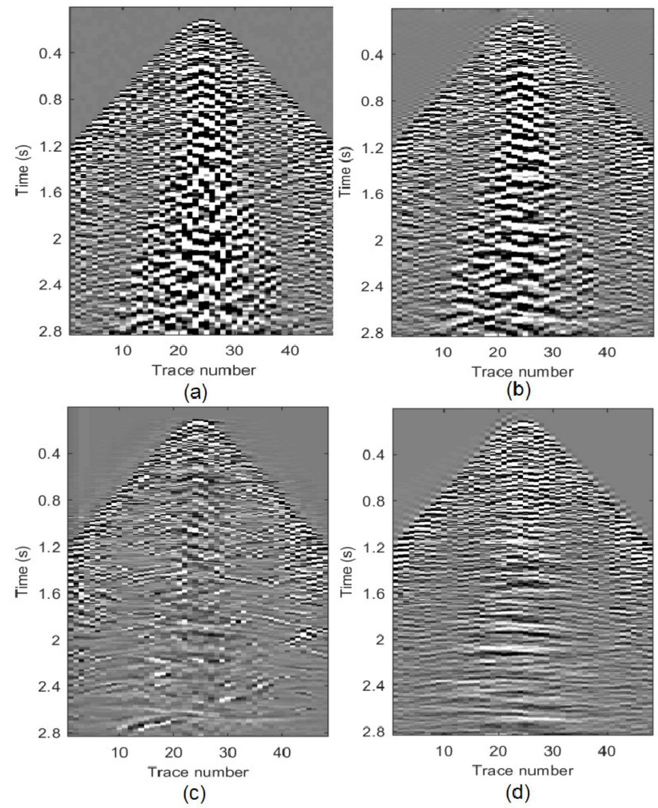

Fig. 16. Shot gather after trace removal and ground-roll attenuation results. (a) Shot gather in Fig. 13 after trace removal. Results after (b) f–k filtering, (c) single CNN, and (d) DFB-CNN. Compare with Figs. 13 and 14.

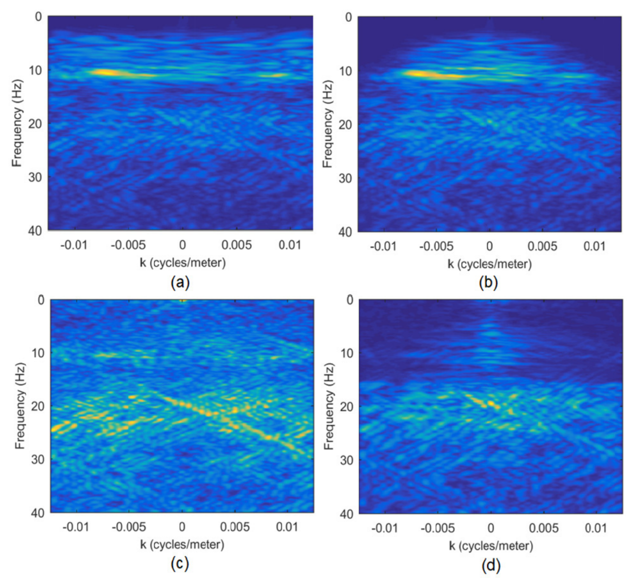

Fig. 17. f–k spectra of the strongly aliased shot gather [Fig. 16(a)] and the ground-roll attenuation results in Fig. 16 by different methods. (a) f–k spectrum of the decimated shot gather [Fig. 16(a)]. The f–k spectra after (b) f–k filtering, (c) single CNN, and (d) DFB-CNN. Compare with Fig. 15.

IV. Discussion

Ground-roll removal is challenging because of: 1) overlapping signal content in the temporal, spatial, and spectral domains; 2) the presence of spatially aliased signal and noise; and 3) the presence of both broad- and narrow-scale features. We proposed a supervised learning strategy to address these challenges. The success of the proposed strategy is due to: 1) the dual-filter bank design to better recognize broad- and narrow-band features and 2) a focusing transform to compact ground-roll and reflection events.

To better demonstrate the proposed method also benefits from the RT transform, we show the denoised result of the shot gather (Fig. 13) using DFB-CNN without the RT transform in Fig. 18. The dual-filter bank concept still does a better

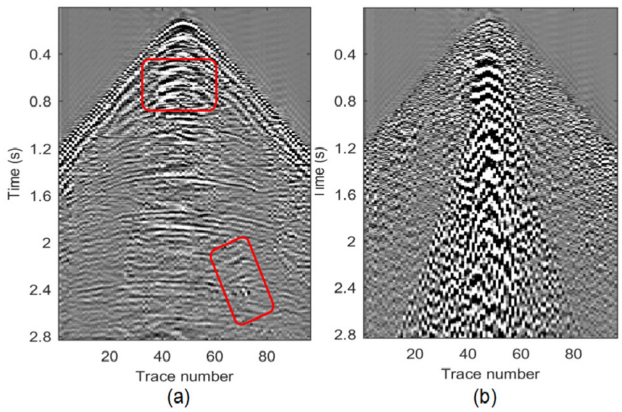

Fig. 18. (a) Ground-roll removal result and (b) difference profile of Fig. 13 using the DFB-CNN without the RT transform.

job than the single CNN [Fig. 14(c) and (d)] for ground-roll attenuation. But there are some residual parts of ground-roll in the top rectangle and ground-roll artifacts in the bottom rectangle (Fig. 18), which is not as good as the result achieved by the proposed method [Fig. 14(e) and (f)]. Thus, the RT transform facilitates separation of the reflections and ground-roll as well.

The dual-filter bank concept for CNNs is related to the principal rationale underlying the use of wavelet transforms in time–frequency analysis. A wavelet transform analyzes high- and low-frequency components using, respectively, short- and long-window lengths, instead of a fixed length as in the short-term Fourier transform [35]–[37]. The use of variable-window lengths for time–frequency analysis is appropriate because of the inverse relationship between spectral and temporal resolutions. Likewise, the networks with variable kernel sizes performed better than a single network with the fixed kernel size. Therefore, we adopted the dual-filter bank approach in this article.

We believe this concept can also be used in other research areas or tasks. For instance, natural images are also characterized by coarse and detailed components [38], [39]. Thus, filter banks of CNNs with different kernel sizes may aid in feature recognition and extraction tasks [40]–[42]. Network complexity tends to increase with the size of the convolutional kernels. A possible solution to tackle this likely undesirable effect is the use of spread-out convolutional kernels that undersample the input image, where the convolutional kernel reads, for instance, every other sample in a row or column, thereby reducing the number of free parameters. This strategy was implemented also in some of the first convolutional networks [43], [44].

Furthermore, the use of a focusing transform, such as the RT transform, to compact signals is also beneficial since it makes it easier for a network with fixed kernel sizes to recognize features of interest, since these features typically extend beyond the size of the convolutional kernel.

V. Conclusion

We propose a DFB-CNN to attenuate ground-roll. In the proposed method, two networks with different architectures and complexities are trained to extract broad- and narrow-scale features separately, which helps with features with dimensions extending beyond the size of the convolutional kernels. The RT transform is added to compact signal and noise for an enhanced feature recognition and extraction. We employ the proposed filter bank implementation of a CNN for ground-roll attenuation as an example of overlapping signal and noise over a range of scales. Tests on both synthetic and real datasets show the superiority of the proposed network in ground-roll attenuation and signal recovery compared with f–k filtering and a single CNN.

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